0/3 -- ESP32 Internal Temperature Sensor

1/3 – Blynk 2.0

2/3 – IFTTT to Google Sheet and LINE

3/3 – Direct to Google Sheet

- ทดสอบส่งค่ากลุ่มตัวแปรจำนวน 10 ค่า ทุกๆ 60 วินาทีไปยัง Google spread sheet
- ค่า 10 จำนวน ได้แก่
    - Item, Data_Time, WiFi_Name, MAC_Address, Internal_ESP32_Tempperature
    - ค่าตัวแปรมาจากการสุ่มแบบจำนวนเต็ม 3 จำนวน ระหว่าง 0-299, 400-599, 700-999
    - ค่าตัวแปรมา DHT22 แบบทศนิยม 2 ตำแหน่ง Temperature, Humidity

Method A: IFTTT

- https://www.allnewstep.com/article/365/วิธีเซฟข้อมูล-arduino-nodemcu-ลง-google-sheet-แบบเรียวไทม์-arduino-esp8266-ifttt-google-sheet
- https://electropeak.com/learn/sending-data-from-esp32-or-esp8266-to-google-sheets-2-methods/#google_vignette

Method B: Google Form

- http://shahrulnizam.com/esp32-lesson-google-sheets-using-google-forms/
- https://randomnerdtutorials.com/esp32-http-get-post-arduino/

Method C: Google Script

- https://iotdesignpro.com/articles/esp32-data-logging-to-google-sheets-with-google-scripts
- https://randomnerdtutorials.com/esp32-datalogging-google-sheets/
- https://docs.leafony.com/en/docs/examples/advanced/2_p/esp32/esp32_googlesheets_1/
- https://randomnerdtutorials.com/esp32-datalogging-google-sheets/
- https://electropeak.com/learn/sending-data-from-esp32-or-esp8266-to-google-sheets-2-methods/#google_vignette

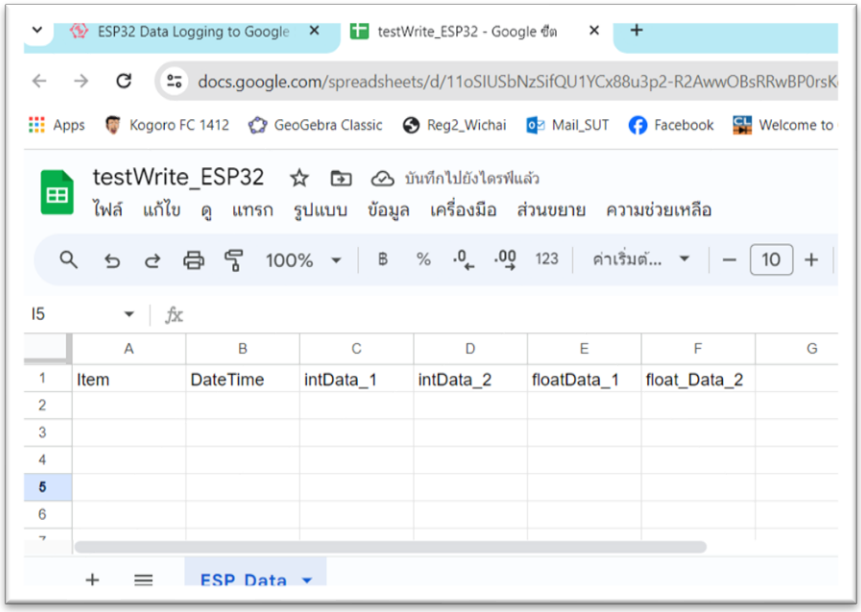
Method D: Google Sheet API

- https://www.praphas.com/forum/index.php?topic=355.0
- https://github.com/mobizt/ESP-Google-Forms-Client

3a/3 – Direct to Google Sheet with Goggle Script

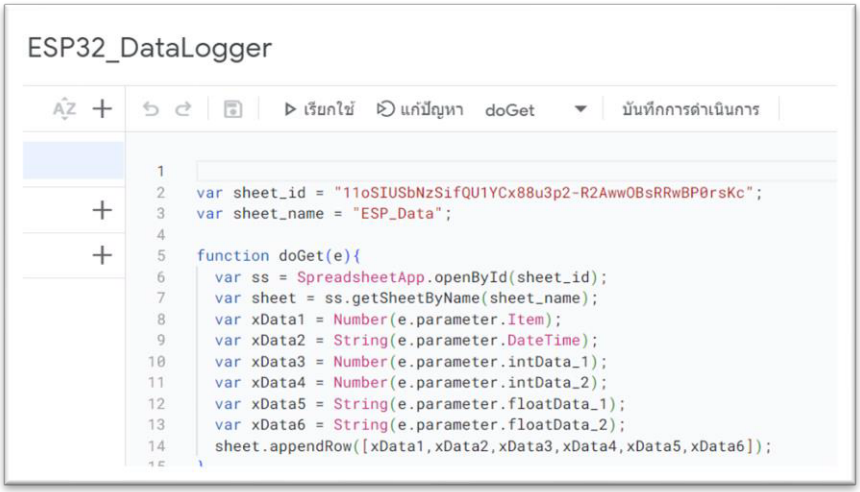
- https://iotdesignpro.com/articles/esp32-data-logging-to-google-sheets-with-google-scripts

1. Create Google Spreadsheet
   - Column = 5
   - Sheet ID = 11oSIU****
   - Sheet Name = ESP_Data



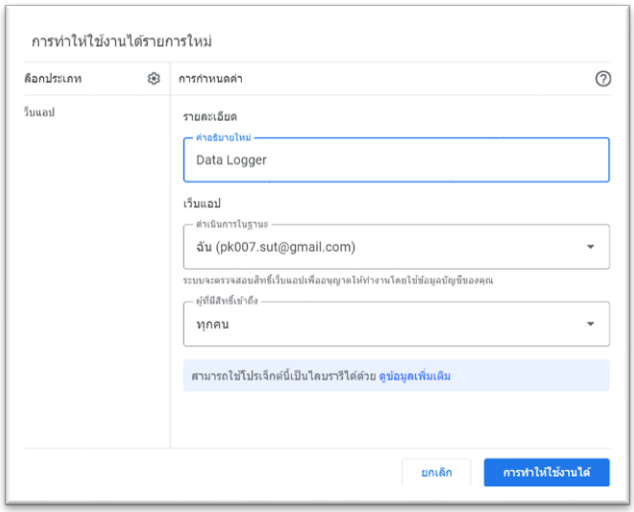
2. Open App Scrip
   - ส่วนขยาย → App Scrips

3. Edit Google Scrip
   - Edit sheet_ID
   - Edit sheet_Name




```
var sheet_id = "11oSIUSbzSifQU1Yx88u3p2-R2AwwOxxxxxxxxx";
var sheet_name = "ESP_Data";

function doGet(e){
 var ss = SpreadsheetApp.openById(sheet_id);
 var sheet = ss.getSheetByName(sheet_name);
 var xData1 = Number(e.parameter.Item);
 var xData2 = String(e.parameter.DateTime);
 var xData3 = Number(e.parameter.intData_1);
 var xData4 = Number(e.parameter.intData_2);
 var xData5 = String(e.parameter.floatData_1);
 var xData6 = String(e.parameter.floatData_2);
 sheet.appendRow([xData1,xData2,xData3,xData4,xData5,xData6]);
}
```

4. Deploy
   - การทำให้ใช้งานได้ → การทำให้ใช้งานได้รายการใหม่
   - เลือดประเภท = เว็บแอป
   - เติมค่า
   - การทำให้ใช้งานได้
   - Copy เวปแอป URL



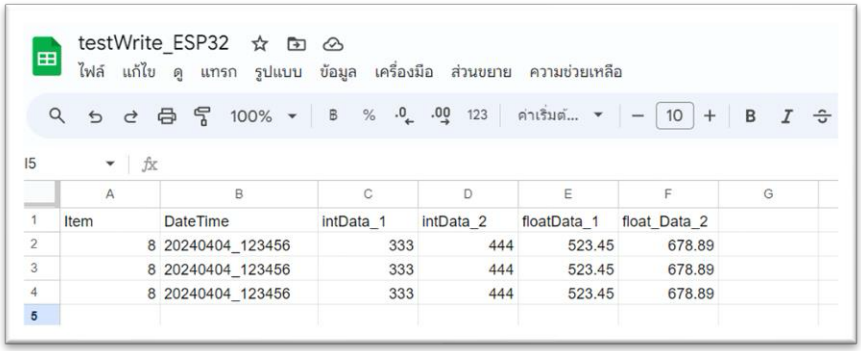
5. เวปแอป URL

https://script.google.com/macros/s/AKfycbyLi1N2xldrwhBOYPHQK5rQCuB_UM_VXq9CFTO /exec

6. ทดสอบส่งผ่าน Browser

https://script.google.com/macros/s/ AKfycbyLi1N2xldrwhBOYPHQK5rQCuB_UM_VXq9CFTO /exec?Item=8&DateTime=20240404_123456&intData_1=333&intData_2=444&floatData_1=523.45&floatData_2=678.89

7. Data to Google Spreadsheet

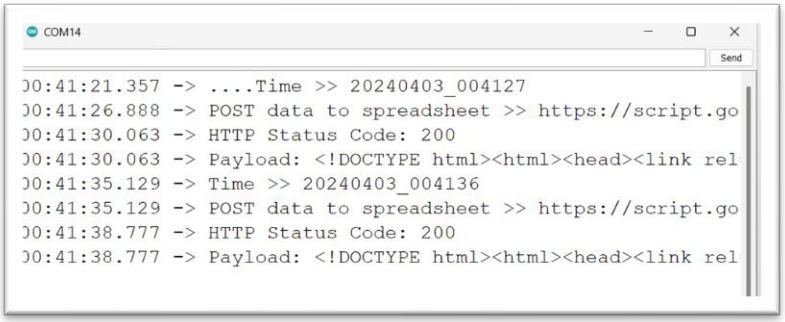


8. ESP32 Code
   - WiFi SSID
   - WiFi Pass
   - Google Scrip ID

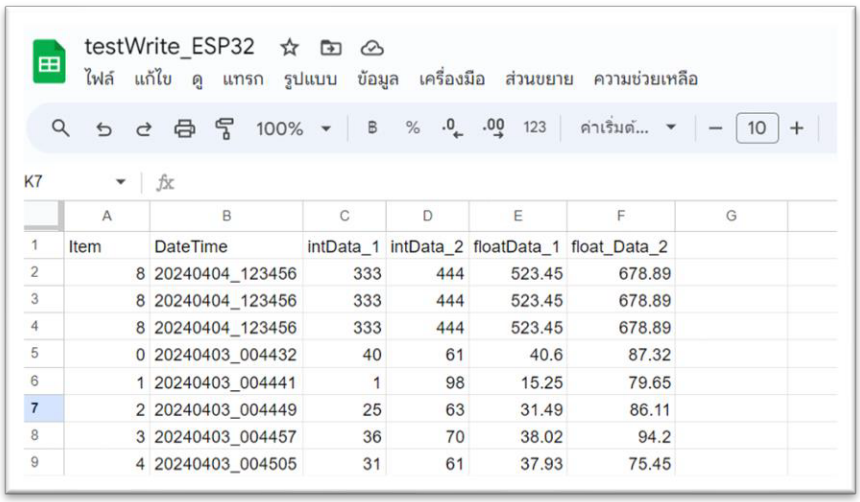
```
#include "WiFi.h"
#include <HTTPClient.h>
#include "time.h"
const char* ntpServer = "pool.ntp.org";
const long  gmtOffset_sec = 25200; // +7*60*60
const int   daylightOffset_sec = 0;

const char* ssid     = "Test1234";
const char* password = "0816601929";

// Google script ID and required credentials
String GOOGLE_SCRIPT_ID = "AKfycbxR3jVNQkEgNU9t4x1tkRvXDhbc2O5ANBaFiAoFCduq";   // change Gscript ID
char timeStringBuff[50]; // 50 chars should be enough for YYYYMMDD-HHMMSS
int nCounter = 0;

void setup() {
  delay(1000);
  Serial.begin(115200);
  Serial.println();
  Serial.print("Connecting to wifi: ");
  Serial.println(ssid);
  Serial.flush();
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  // Init and get the time
  configTime(gmtOffset_sec, daylightOffset_sec, ntpServer);
}
```

```
void loop() {
 if (WiFi.status() == WL_CONNECTED) {
  static bool flag = false;
  struct tm timeinfo;
  if (!getLocalTime(&timeinfo)) {
    Serial.println("Failed to obtain time");
    return;
  }
  strftime(timeStringBuff, sizeof(timeStringBuff), "%Y%m%d_%H%M%S", &timeinfo);
  String asString(timeStringBuff);
  Serial.print("Time >> ");
  Serial.println(asString);

  String urlFinal = "https://script.google.com/macros/s/" + GOOGLE_SCRIPT_ID + "/exec?" +
              "Item=" + String(nCounter) +
              "&DateTime=" + asString +
              "&intData_1=" + String(random(49)) +
              "&intData_2=" + String(random(61, 99)) +
              "&floatData_1=" + String(random(199, 4999) / 100.0) +
              "&floatData_2=" + String(random(6999, 9999) / 100.0);

  Serial.print("POST data to spreadsheet >> ");
  Serial.println(urlFinal);
  HTTPClient http;
  http.begin(urlFinal.c_str());
  http.setFollowRedirects(HTTPC_STRICT_FOLLOW_REDIRECTS);
  int httpCode = http.GET();
  Serial.print("HTTP Status Code: ");
  Serial.println(httpCode);
  //--------------------------------------------------------------------------
  //getting response from google sheet
  String payload;
  if (httpCode > 0) {
    payload = http.getString();
    Serial.println("Payload: " + payload);
  }
  //--------------------------------------------------------------------------
  http.end();
 }
 nCounter++;
 delay(5000);
}
```

9. Run



10. Result



	A	B	C	D	E	F	G
1	Item	DateTime	intData_1	intData_2	floatData_1	float_Data_2	
2	8	20240404_123456	333	444	523.45	678.89	
3	8	20240404_123456	333	444	523.45	678.89	
4	8	20240404_123456	333	444	523.45	678.89	
5	0	20240403_004432	40	61	40.6	87.32	
6	1	20240403_004441	1	98	15.25	79.65	
7	2	20240403_004449	25	63	31.49	86.11	
8	3	20240403_004457	36	70	38.02	94.2	
9	4	20240403_004505	31	61	37.93	75.45	